

WIPRO TALENTNEXT PROGRAM

PBL HANDS-ON ASSIGNMENT

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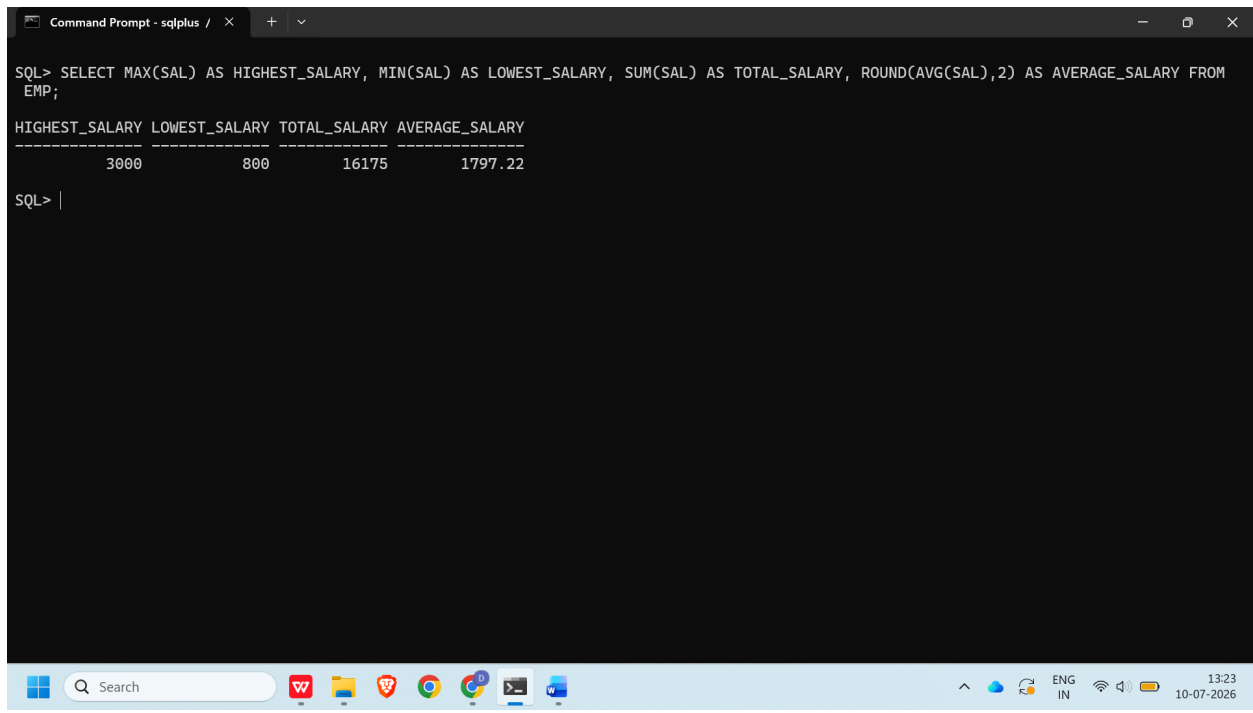
Topic: Reporting Aggregated Data Using the Group Functions

Question 1

Find the highest, lowest, sum, and average salary of all employees, Label the columns.

Query:

```
SELECT MAX(SAL) AS HIGHEST_SALARY, MIN(SAL) AS LOWEST_SALARY, SUM(SAL) AS TOTAL_SALARY, ROUND(AVG(SAL),2) AS AVERAGE_SALARY FROM EMP;
```



The screenshot shows a Windows Command Prompt window titled "Command Prompt - sqlplus /". The prompt is "SQL>". The user has entered the query: "SELECT MAX(SAL) AS HIGHEST_SALARY, MIN(SAL) AS LOWEST_SALARY, SUM(SAL) AS TOTAL_SALARY, ROUND(AVG(SAL),2) AS AVERAGE_SALARY FROM EMP;". The output is displayed as a table with four columns: HIGHEST_SALARY, LOWEST_SALARY, TOTAL_SALARY, and AVERAGE_SALARY. The values are 3000, 800, 16175, and 1797.22 respectively. The prompt is now "SQL> |".

```
SQL> SELECT MAX(SAL) AS HIGHEST_SALARY, MIN(SAL) AS LOWEST_SALARY, SUM(SAL) AS TOTAL_SALARY, ROUND(AVG(SAL),2) AS AVERAGE_SALARY FROM EMP;
```

HIGHEST_SALARY	LOWEST_SALARY	TOTAL_SALARY	AVERAGE_SALARY
3000	800	16175	1797.22

```
SQL> |
```

Question 2

Maximum, Minimum, Sum, and Average, respectively. Round your results to the nearest whole number.

Query:

```
SELECT ROUND(MAX(SAL)) AS Maximum, ROUND(MIN(SAL)) AS Minimum,  
ROUND(SUM(SAL)) AS Sum, ROUND(AVG(SAL)) AS Average FROM EMP;
```

Output:

```
SQL> SELECT ROUND(MAX(SAL)) AS Maximum, ROUND(MIN(SAL)) AS Minimum, ROUND(SUM(SAL)) AS Sum, ROUND(AVG(SAL)) AS Average FROM EMP;  
  
MAXIMUM    MINIMUM    SUM    AVERAGE  
-----  
3000       800      16175    1797
```

Question 3

Modify the above query to display the minimum, maximum, sum, and average salary for each job type.

Query:

```
SELECT JOB, ROUND(MIN(SAL)) AS Minimum, ROUND(MAX(SAL)) AS Maximum,  
ROUND(SUM(SAL)) AS Sum, ROUND(AVG(SAL)) AS Average FROM EMP GROUP BY JOB;
```

Output:

```
SQL> SELECT JOB, ROUND(MIN(SAL)) AS Minimum, ROUND(MAX(SAL)) AS Maximum, ROUND(SUM(SAL)) AS Sum, ROUND(AVG(SAL)) AS Average FROM EMP  
GROUP BY JOB;  
  
JOB          MINIMUM    MAXIMUM    SUM    AVERAGE  
-----  
CLERK         800        950      1750     875  
SALESMAN     1250       1600     5600    1400  
ANALYST      3000       3000     3000    3000  
MANAGER     2850       2975     5825    2913
```

Question 4

Write a query to display the number of people with the same job.

Query:

```
SELECT JOB, COUNT(*) AS TOTAL_EMPLOYEES FROM EMP GROUP BY JOB;
```

Output:

```
SQL> SELECT JOB, COUNT(*) AS TOTAL_EMPLOYEES FROM EMP GROUP BY JOB;
```

JOB	TOTAL_EMPLOYEES
CLERK	2
SALESMAN	4
ANALYST	1
MANAGER	2

```
SQL >
```

Question 5

Determine the number of managers without listing them. Label the column Number of Managers.

Query:

```
SELECT COUNT(DISTINCT MGR) AS "Number of Managers" FROM EMP;
```

Output:

```
SQL> SELECT COUNT(DISTINCT MGR) AS "Number of Managers" FROM EMP;
```

Number of Managers
4

Question 6

Find the difference between the highest and lowest salaries. Label the column DIFFERENCE.

Query:

```
SELECT MAX(SAL)-MIN(SAL) AS DIFFERENCE FROM EMP;
```

Output:

```
SQL> SELECT MAX(SAL)-MIN(SAL) AS DIFFERENCE FROM EMP;

DIFFERENCE
-----
        2200
```

Question 7

Create a report to display the manager number and the salary of the lowest-paid employee for that manager. Exclude anyone whose manager is not known. Exclude any groups where the minimum salary is \$2000 or less. Sort the output in descending order of salary.

Query:

```
SELECT MGR, MIN(SAL) AS LOWEST_SALARY FROM EMP WHERE MGR IS NOT NULL
GROUP BY MGR HAVING MIN(SAL) > 2000 ORDER BY LOWEST_SALARY DESC;
```

Output:

```
SQL> SELECT MGR, MIN(SAL) AS LOWEST_SALARY FROM EMP WHERE MGR IS NOT NULL GROUP BY MGR HAVING MIN(SAL) > 2000 ORDER BY LOWEST_SALARY
DESC;

MGR  LOWEST_SALARY
-----
7566          3000
7839          2850

SQL>
```

Question 8

Create a query to display the total number of employees and, of that total, the number of employees hired in 1980, 1981, and 1982. Create appropriate column headings.

Query:

```
SELECT COUNT(*) AS TOTAL_EMPLOYEES,
COUNT(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1980',1)) AS "1980",
COUNT(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1981',1)) AS "1981",
COUNT(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1982',1)) AS "1982" FROM EMP;
```

Output:

```
Command Prompt - sqlplus / X + v

SQL> SELECT COUNT(*) AS TOTAL_EMPLOYEES, SUM(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1980',1,0)) AS "1980", SUM(DECODE(TO_CHAR(HIREDATE,'YYYY'), '1981',1,0)) AS "1981", SUM(DECODE(TO_CHAR(HIREDATE,'YYYY'), '1982',1,0)) AS "1982" FROM EMP;

TOTAL_EMPLOYEES      1980      1981      1982
-----
          9             1             7             0

SQL> SELECT COUNT(*) AS TOTAL_EMPLOYEES, COUNT(DECODE(TO_CHAR(HIREDATE,'YYYY'),'1980',1)) AS "1980", COUNT(DECODE(TO_CHAR(HIREDATE,'YYYY'), '1981',1)) AS "1981", COUNT(DECODE(TO_CHAR(HIREDATE,'YYYY'), '1982',1)) AS "1982" FROM EMP;

TOTAL_EMPLOYEES      1980      1981      1982
-----
          9             1             7             0

SQL> |
```

